Thermodynamic Dynamics of Observer-Memory Entanglement

Memory, formed self-frames and revisable access

Complete Collected Volume

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Introduction

How can retained experience become an organization through which a system meets its next task, and then revises that organization? T-DOME develops the memory–self–introspection route through different but connected questions of task information, formation and continued self-correction.

I prices the usable correlations in an acquired history and the gross feedback work lost by compressing its posterior. II gives a finite formation theorem: training on an unknown task–channel coupling selects a two-bit reading frame that beats every history-independent direction, even with an oracle regression coefficient for the latter. Both use one test read. The coefficient oracle is granted free; retained memory and total cost are not matched. This proves the value of history-selected access, not a distinction from general learned parameters. A correct static frame can remain useful when frozen; formation does not require perpetual change.

III studies diagnosis and decisions when the interface or body changes. Even residuals can reveal drift without its sign. The same acquired joint belief can authorize a task, value another inquiry and select paid repair; equal sign confidence need not imply the same repair. A terminal sufficient statistic need not support the next update. The finite formation example and the drifting body have different supplied dynamics, so the trilogy does not turn them into a universal necessity chain or a proof of experience.

Reading map

TD-I Task-relevant retained distinctions have usable work value. Compression prices a current feedback posterior, not every future learning task.

TD-II Finite paid training selects a two-bit reading frame that beats a history-independent direction at one test read, even with a free coefficient oracle. Total resources are not matched; the independent-replay control tests history relevance, not a separation from general learning.

TD-III Drift detection, inquiry and paid repair use different functions of the same acquired belief. Terminal compression and continuing self-revision need not retain the same information.